

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 31 January 2025

Team ID: LTVIP2025TMID35065

Project Name: Trafficelligence – Intelligent Traffic Volume Prediction System

Maximum Marks: 4 Marks

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web dashboard and mobile-friendly interface	React.js, Bootstrap, HTML/CSS
2	Application Logic-1	Handles user authentication and session management	Python (Flask/Django)
3	Application Logic-2	Processes traffic data ingestion and cleaning	Python (Pandas, ETL scripts)
4	Application Logic-3	Executes traffic prediction via ML model	Python (Scikit-Learn, XGBoost server)
5	Database	Structured storage for users, predictions, logs	PostgreSQL
6	Cloud Database	Scalable cloud analytics storage	AWS RDS (PostgreSQL)
7	File Storage	Stores uploaded datasets, model artifacts	AWS S3
8	External API-1	Fetches weather and traffic sensor data	OpenWeatherMap API
9	External API-2	(Optional) For integration with external data sources	Google Maps API
10	Machine Learning Model	Predicts traffic volumes	XGBoost regression
11	Infrastructure	Application hosting and orchestration	AWS EC2, EKS (Kubernetes), Docker

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Web app, ETL, model training, dashboard	React.js, Flask, Pandas, Scikit-Learn, XGBoost

S.No	Characteristics	Description	Technology
2	Security Implementations	Secure access, data protection, firewall, authentication	HTTPS/TLS, JWT, OAuth2, AWS IAM, OWASP best practices
3	Scalable Architecture	Modular microservices for web, ML engine, ingestion	Docker, Kubernetes (EKS), AWS Auto Scaling
4	Availability	High availability and failover	AWS Load Balancer, Multi-AZ deployment, S3 backup
5	Performance	Fast response and efficient data handling	Redis cache, tuned DB indexes, CDN for static content
